

Measurement and Conversion

Introduction: When Math Takes You Places

Imagine waking up in Manila, having breakfast, and then video-calling a friend in New York who is just about to eat dinner. Later that day, you book a flight to Tokyo, convert your pesos into yen, and use a map to figure out how far your hotel is from Mount Fuji. None of this would be possible without three essential mathematical tools: **map scales**, **currency conversion**, and **time zone conversion**.

Every day, travelers, businesspeople, overseas Filipino workers (OFWs), and even online shoppers use these skills. A map scale tells you that 1 centimeter on paper might equal 10 kilometers on the road. Currency conversion helps you know that your ₱1,000 is worth about \$17 USD. Time zone conversion lets you call your loved ones abroad without waking them up in the middle of the night.

This reading material examines how to interpret and use map scales to calculate real-world distances, convert currencies using exchange rates, and convert time across different time zones. These skills are not just abstract math problems — they are practical tools for making smart travel decisions, budgeting for trips, and staying connected with the world.

Part 1: Maps and Scales – From Paper to Reality

What Is a Map Scale?

A **map scale** shows the relationship between a distance measured on a map and the actual distance on the ground. Without a scale, a map is just a drawing — with a scale, it becomes a reliable tool for planning travel.

There are three common ways to express map scales:

Type	Example	Meaning
Ratio Scale (Representative Fraction)	1:100,000	1 unit on map = 100,000 of the same units in real life
Bar Scale (Graphic Scale)	A line marked with kilometers	Visually shows how map distance translates to real distance
Verbal Scale	"1 cm = 10 km"	Direct statement in words

How to Convert Map Distance to Real Distance

Step 1: Measure the distance on the map using a ruler (in centimeters or inches).

Step 2: Identify the scale (e.g., 1:1,000,000 or 1 cm = 10 km).

Step 3: Multiply the map distance by the real-world equivalent.

Example 1 (Ratio Scale):

You measure 22 cm between Manila and Baguio on a map with scale 1:1,000,000.

- 1:1,000,000 means 1 cm = 10 km (since $1,000,000 \text{ cm} \div 100,000 = 10 \text{ km}$)
- Real distance = $22 \text{ cm} \times 10 \text{ km/cm} = \mathbf{220 \text{ km}}$

Example 2 (Verbal Scale):

A hotel is 3 cm from the Eiffel Tower on a map where 1 cm = 50 m.

- Real distance = $3 \text{ cm} \times 50 \text{ m/cm} = \mathbf{150 \text{ m}}$ (walking distance)

Example 3 (Fractional Scale):

A cruise map shows Manila to Ho Chi Minh City as 16 cm with scale 1:10,000,000.

- $1 \text{ cm} = 10,000,000 \text{ cm} \times 1 \text{ km}/100,000 \text{ cm} = 100 \text{ km}$
 - Real distance = $16 \times 100 \text{ km} = \mathbf{1,600 \text{ km}}$
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Part 2: Currency Conversion – Understanding the Value of Money

What Is Currency Conversion?

Currency is the system of money used in a particular country (e.g., Philippine Peso, US Dollar, Euro, Japanese Yen). **Currency conversion** is the process of changing one currency into another using an **exchange rate**.

An **exchange rate** tells you how much one currency is worth in terms of another. For example, if $1 \text{ USD} = 59 \text{ PHP}$, then one US dollar can be exchanged for 59 Philippine pesos.

Key Terms in Currency Conversion

Term	Definition
Base Currency	The first currency listed in a pair (e.g., USD in USD/PHP)
Quote Currency	The second currency in the pair (e.g., PHP in USD/PHP)
Exchange Rate	The value of one currency in terms of another
Forex (Foreign Exchange)	The global market where currencies are traded

How to Convert Currencies

To convert from Base Currency to Quote Currency (e.g., USD to PHP):

Multiply by the exchange rate.

Amount in PHP = Amount in USD $\times$ Exchange Rate

To convert from Quote Currency to Base Currency (e.g., PHP to USD):

Divide by the exchange rate.

Amount in USD = Amount in PHP ÷ Exchange Rate

Example 1 (USD to EUR):

Nicole has 500 USD. Exchange rate: 1 USD = 0.88 EUR.

Euros = $500 \times 0.88 = \mathbf{440 \text{ EUR}}$

Example 2 (PHP to USD):

Mr. De La Cruz has ₱100,000. Exchange rate: 1 USD = 58.50 PHP.

USD = $100,000 \div 58.50 = \mathbf{\$1,709.40}$

Example 3 (USD to PHP for online shopping):

A dress costs 150 USD. Exchange rate: 1 USD = 57 PHP.

Price in PHP = $150 \times 57 = \mathbf{₱8,550}$

Part 3: Time Zone Conversion – Synchronizing the World

What Are Time Zones?

A **time zone** is a **geographic** region of the Earth that follows a uniform standard time. Since the Earth rotates 15° every hour, there are 24 time zones, each roughly 15° of longitude wide. The reference point is the **UTC (Coordinated Universal Time)**, also known as GMT (Greenwich Mean Time).

The Philippines is in **UTC+8** (8 hours ahead of UTC). When it is noon in London (UTC+0), it is 8:00 PM in Manila.

Key Terms in Time Conversion

Term	Definition
UTC (Coordinated Universal Time)	The global time standard
UTC Offset	The difference in hours between a time zone and UTC
Daylight Saving Time (DST)	Seasonal adjustment moving clocks forward 1 hour

Term	Definition
Local Time	The actual time observed in a specific place

How to Convert Time Between Zones

Step 1: Identify the UTC offset of both locations.

Step 2: Find the difference (add if going east, subtract if going west).

Step 3: Apply the difference to the given time.

Example 1 (Manila to New York during DST):

Manila (UTC+8), New York (UTC-4 during DST). Difference = 12 hours (Manila is ahead).

If interview is 10:00 AM ET in New York:

Manila time = 10:00 AM + 12 hours = **10:00 PM PHT (same day)**

Example 2 (Manila to Dubai):

Manila (UTC+8), Dubai (UTC+4). Philippines is 4 hours ahead.

If Lea calls at 7:00 PM PHT:

Dubai time = 7:00 PM - 4 hours = **3:00 PM GST**

Example 3 (Seoul to Bangkok):

Seoul (UTC+9), Bangkok (UTC+7). South Korea is 2 hours ahead.

If Criselda calls at 8:00 PM KST:

Bangkok time = 8:00 PM - 2 hours = **6:00 PM ICT**

Example 4 (Flight arrival with time zone adjustment):

Flight departs New York (UTC-5) at 6:00 PM local time. Flight duration = 10 hours.

Destination: Paris (UTC+1).

Arrival in UTC = (6:00 PM - 5 hours to UTC) + 10 hours = 11:00 PM UTC.

Convert to Paris time: 11:00 PM UTC + 1 hour = **12:00 AM (midnight) next day in**

Paris

Part 4: Putting It All Together – Travel Scenarios

Scenario 1: Road Trip in the Philippines

You are planning a road trip from Manila to Baguio. The map scale is 1:1,000,000. You measure the distance as 25 cm.

- Real distance = $25 \text{ cm} \times 1,000,000 \times 1\text{km}/100,000\text{cm} = \mathbf{250 \text{ km}}$

Scenario 2: Budgeting for a Trip to Japan

Your budget is US\$1,200. Exchange rate: 1 USD = 145.50 JPY. Bank charges 2% conversion fee.

- JPY before fee = $1,200 \times 145.50 = 174,600 \text{ JPY}$
- Fee = $174,600 \times 0.02 = 3,492 \text{ JPY}$
- Final amount = $174,600 - 3,492 = \mathbf{171,108 \text{ JPY}}$

Scenario 3: Virtual Team Meeting Across Continents

Team members in: Manila (UTC+8), Paris (UTC+2 during DST), New York (UTC-4 during DST). Meeting at 4:00 PM Paris time.

- Manila: 4:00 PM + 6 hours = **10:00 PM PHT**
- New York: 4:00 PM - 6 hours = **10:00 AM EDT**

Part 5: Common Conversion Factors (Reference Table)

Length

1 cm on map (scale 1:100,000)	1 km in real life
1 inch on map (scale 1:63,360)	1 mile in real life

Currency (Sample Rates - for practice only)

1 USD	≈ 59 PHP
1 EUR	≈ 71 PHP
1 USD	≈ 0.85 EUR
1 USD	≈ 159 JPY

Time Zones (Common Offsets)

Philippines (PHT)	UTC+8
Japan (JST)	UTC+9
Dubai (GST)	UTC+4
Paris (CET)	UTC+1 (UTC+2 during DST)
New York (EST)	UTC-5 (UTC-4 during DST)

Key Terms

Term	Definition
Map Scale	The relationship between distance on a map and actual distance on the ground
Ratio Scale	A numerical scale written as a ratio (e.g., 1:100,000)
Bar Scale	A visual scale shown as a line marked with distances
Currency	The system of money used in a particular country
Exchange Rate	The value of one currency in terms of another
Base Currency	The first currency listed in a currency pair
Quote Currency	The second currency listed in a currency pair
Time Zone	A region following a standard time
UTC (Coordinated Universal Time)	The global time standard
UTC Offset	The difference in hours between a time zone and UTC
Daylight Saving Time (DST)	Seasonal adjustment moving clocks forward 1 hour

Review Questions

Write your answers on a separate sheet of paper.

1. A map has a scale of 1:500,000. If two cities are 8 cm apart on the map, what is the actual distance in kilometers?
2. You are traveling from Manila (UTC+8) to London (UTC+0). Your flight departs Manila at 9:00 PM and takes 14 hours. What time will you arrive in London?
3. A laptop costs \$650 USD. The exchange rate is 1 USD = 56 PHP. How much is the laptop in Philippine pesos?
4. On a map with scale 1 cm = 25 km, you measure a distance of 4.5 cm. What is the real-world distance?
5. You have €200 Euros. The exchange rate is 1 EUR = 1.10 USD. How many US dollars will you receive?
6. A virtual meeting is scheduled at 2:00 PM in New York (UTC-5). What time is the meeting in Manila (UTC+8)?
7. A map scale is written as 1:250,000. Convert this to a verbal scale in kilometers.
8. An OFW in Japan sends ¥50,000 to the Philippines. The exchange rate is 1 PHP = 2.50 JPY. How much is the remittance in Philippine pesos?
9. Your flight departs Tokyo (UTC+9) at 8:00 AM and arrives in Dubai (UTC+4) after 10 hours. What is the local arrival time in Dubai?
10. Why is it important to understand map scales, currency conversion, and time zones when planning international travel? Give one example for each.

Answer Key

1. Actual distance = $8 \text{ cm} \times (500,000 \div 100,000 \text{ cm/km}) = 40 \text{ km}$
2. Arrival time in London = 3:00 AM next day in London
3. Price in PHP = $\text{USD } 650 \times \text{₱}56/\text{USD} = \text{₱}36,400$
4. Real distance = $4.5 \text{ cm} \times 25 \text{ km/cm} = 112.5 \text{ km}$
5. USD received = $200 \text{ EUR} \times 1.10 \text{ USD/EUR} = 220 \text{ USD}$
6. Manila time = 3:00 AM next day
7. $1 \text{ cm} = 2.5 \text{ km}$
8. PHP received = $50,000 \text{ JPY} \div 2.50 \text{ JPY / PHP} = 20,000 \text{ PHP}$
9. Arrival in Dubai = 1:00 PM local time